

Aggregate Benchmarks Are an Unreliable Map of Capability: Why Distribution-Matched Evaluation Must Precede Data Investment

Perpetual Labs Research

Perpetual Labs — Applied AI Research

Research Note PL-RN-2026-i08 · correspondence: research@perpetual-labs.ai

Abstract

Two separately-argued literatures converge on a single uncomfortable conclusion: aggregate, headline benchmarks (MMLU-style accuracy, BIG-Bench aggregates) cannot reliably tell you where a model is weak — and therefore where data is needed. The first thread is the emergence debate: Schaeffer et al. show that over 92% of claimed "emergent abilities" appear under just two nonlinear metrics (Multiple-Choice Grade, Exact-String-Match), and that swapping to a linear metric (Token Edit Distance) on the **same model outputs** dissolves the discontinuity [1][2]. The second thread is the loss-vs-downstream gap: Isik et al. show that under distribution misalignment, downstream BLEU can **worsen** with more pretraining even while cross-entropy improves monotonically [3], and Ruan et al. show capability lives on a ~3-dimensional manifold (top-3 PCs ≈97% variance) that aggregates obscure [4]. We extend the original briefing into a rigorous synthesis, quantify the failure modes with verbatim numbers (MMLU-Pro drops 16–33% vs MMLU [5]; inference-time decontamination removes 22.9% inflation on GSM8K and 19.0% on MMLU [6]; clinical benchmark accuracy averages 57.2% while MMLU exceeds 88% [10] [11]), corroborate them at the 2025–2026 frontier (the SWE-Bench Illusion memorization audit [15]; contamination-free LiveMedBench, best model 39.2% [17]; GPQA Diamond near-saturated at ~90% while Humanity's Last Exam still only ~37–44% no-tools even at the frontier (44.4% Gemini 3.1 Pro) [20]; ARC-AGI-2 ~3% for frontier models vs ~75% for humans [18]), and derive the operational consequence for a data business: because the measurement problem strictly precedes the data-creation decision, deployment-distribution-matched evaluation is a prerequisite product, not a nicety — and a defensible one. This 57.2%-vs-88% comparison is a contrast across different models and studies on benchmark items — not a controlled head-to-head, and not a measurement on real patients.

1. Introduction

The dominant mental model for improving a language model is a feedback loop: measure capability on a benchmark, find the weak spots, acquire or build data targeting those weak spots, retrain, re-measure. Every step of this loop is only as trustworthy as its first step — the measurement. This paper argues, with quantitative grounding, that the measurement step as commonly practiced is unreliable enough to invalidate the entire loop when the benchmark is an aggregate (a single headline accuracy number averaged over many heterogeneous tasks) and the deployment target is a specialized distribution.

The original i08 briefing made this claim by joining two debates that the field usually treats separately: (a) the 'emergent abilities are a mirage' debate, about whether benchmark jumps are real or metric artifacts [1][2], and (b) the 'does pretraining loss track downstream task quality' debate, about whether lower loss guarantees better task performance [3]. The synthesis is that **neither** aggregate accuracy nor aggregate loss is a faithful guide to where

capability — and therefore data — is lacking, and that the only faithful guide is evaluation matched to the deployment distribution.

We take that briefing and build it into a full argument: we formalize the two failure mechanisms (metric-induced distortion and distribution-induced divergence), quantify each with numbers copied verbatim from primary sources, marshal 2024–2026 corroborating evidence (benchmark saturation, contamination, the clinical knowledge-practice gap), state the assumptions and counter-evidence explicitly, and derive the implication for an organization whose business is creating and selling vertical training data: the eval comes first, and it is itself the scarce, defensible product.

2. Background and Related Work

Wei et al. [2] defined an emergent ability as one 'not present in smaller models but present in larger models,' appearing as a sharp, unpredictable transition. They catalogued such transitions across BIG-Bench and the Massive Multitask (MMLU) benchmark, reporting that GPT-3's performance jumped sharply above random at $\sim 2 \times 10^{22}$ training FLOPs (13B params) and that several abilities emerged for LaMDA at $\sim 10^{23}$ FLOPs (~ 100 B params) [2]. This framing — capability as discontinuous and scale-gated — directly shaped data and compute strategy: if abilities switch on at thresholds, the planning question becomes 'when do we cross the threshold,' not 'where is the distribution thin.'

Schaeffer et al. [1] contested the framing's foundation. Analyzing fixed model outputs, they showed that the discontinuity is largely a property of the **metric**: nonlinear/discontinuous metrics (Exact-String-Match, Multiple-Choice-Grade) manufacture apparent emergence, while linear metrics (Token Edit Distance) on the identical outputs yield smooth, predictable curves. Over 92% of claimed emergent abilities appeared under just those two nonlinear metrics, and of the 39 preferred BIG-Bench metrics at most 5 displayed emergence (4/39 under their hand annotation) [1].

Separately, the scaling-laws literature asked whether pretraining loss predicts downstream performance. Isik et al. [3] showed the answer is conditional on distribution alignment. Ruan et al. [4] reframed the predictability question observationally, fitting scaling laws across ~ 100 public models and finding capability occupies a low-dimensional manifold. The contemporary evaluation-engineering literature (MMLU-Pro [5], contamination studies [6], Humanity's Last Exam [7], LiveBench-style rolling evals [9]) is the practical response to the saturation and contamination of aggregates; the 2025–2026 frontier has hardened this response with memorization audits of headline benchmarks (the SWE-Bench Illusion, showing models locate buggy files from the issue text alone at 76% in-benchmark but only $\sim 53\%$ out-of-benchmark [15]), execution-verified agent suites that frontier models still fail (Terminal-Bench 2.0, $< 65\%$ for frontier models/agents [16]), and contamination-controlled vertical benchmarks (LiveMedBench, best model 39.2% with 84% of models degrading post-cutoff [17]). Insight i09 in this map treats the **temporal** decay of evals; this paper treats the orthogonal **spatial/distributional** failure — aggregates being the wrong map even when uncontaminated.

3. Mechanism I — Metric-Induced Distortion

The first reason aggregates mislead is that the scoring function, not the model, can dominate the shape of the reported capability curve. Schaeffer et al.'s central demonstration is a controlled one: they hold the model's raw outputs fixed and vary only the metric [1]. Under Exact-String-Match on a multi-digit arithmetic task, a model that is off by one digit scores identically to one off by a billion — the metric is geometric in target length, so per-token

improvements are invisible until they all align, producing an apparent 'jump.' Under Token Edit Distance, which is quasi-linear in target length, the same outputs reveal smooth, monotonic improvement [1].

The aggregate-eval relevance is direct. A headline benchmark that averages many sub-tasks scored with such nonlinear metrics will report a model as 'flat then jumping' on a capability that is in fact improving continuously — and will report a *plateau* (zero apparent gain) precisely in the regime where targeted data could be moving the needle sub-threshold. The metric hides the gradient that a data buyer most needs to see. Schaeffer et al. quantify the concentration of the artifact: of the 39 preferred BIG-Bench metrics, at most 5 produce emergence, and over 92% of claimed emergent abilities sit under just two nonlinear metrics (Multiple-Choice-Grade and Exact-String-Match) [1].

The counter-position (Section 9) is that some abilities may genuinely be threshold-like [2]; even granting that, the practical lesson survives: an aggregate score whose movement is governed by metric nonlinearity is not a measurement you can locate a data gap with. You cannot tell 'the model cannot do X' from 'the metric refuses to credit partial progress on X.'

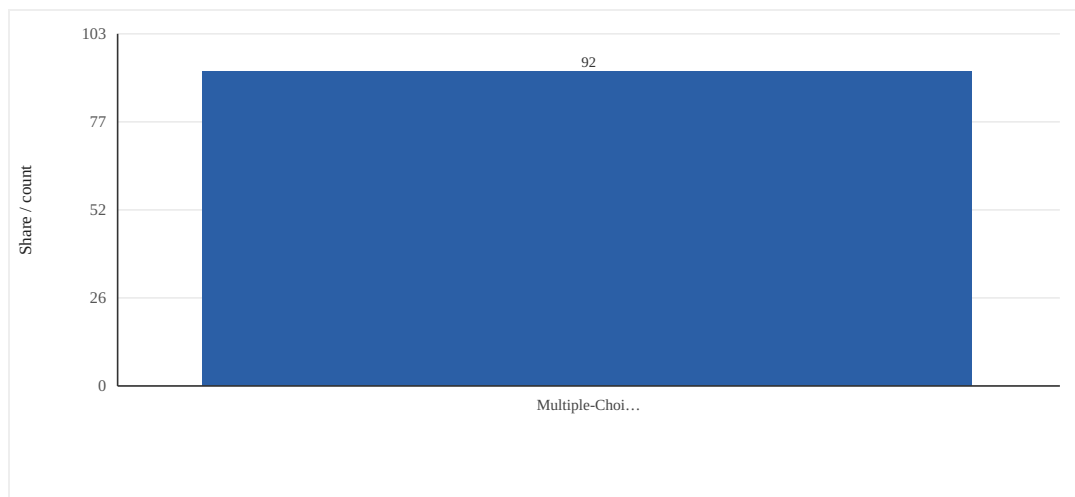


Figure 1. Metric choice manufactures or dissolves 'emergence' (same model outputs). Schaeffer et al. hold model outputs fixed and vary only the metric: over 92% of claimed emergent abilities appear under just two nonlinear metrics (Multiple-Choice-Grade, Exact-String-Match), and of the 39 preferred BIG-Bench metrics at most 5 display emergence (4 under hand annotation) — switching to Token Edit Distance yields smooth, predictable scaling on the identical outputs. Schaeffer, Miranda & Koyejo, *NeurIPS 2023 (arXiv:2304.15004)*, *BIG-Bench meta-analysis*. · Empirical data.

4. Mechanism II — Distribution-Induced Divergence

The second reason is sharper and more damaging to the standard loop: even a perfect, continuous loss metric can move in the *opposite direction* to the task you care about when the pretraining distribution and the deployment distribution are misaligned. Isik et al. [3] establish this for machine translation. Holding finetuning fixed and scaling pretraining tokens, they find that with sufficient alignment both downstream cross-entropy and BLEU improve monotonically, but under moderate misalignment with relatively small finetuning sets the BLEU scaling law breaks — BLEU fluctuates or worsens once the pretraining dataset size passes a threshold while the cross-entropy loss continues to scale as a power law [3]. This is the formal refutation of 'lower loss $\Rightarrow$ better task.'

Loss is an aggregate over the pretraining distribution; the task lives on a different distribution; the map between them is not monotone under shift. A team watching loss (or any pretraining-distribution aggregate) would conclude 'more data is helping' at exactly the moment the deployment metric was degrading.

Ruan et al. [4] supply the structural reason aggregates conceal this: capability is low-dimensional. Across ~100 models, the top 3 principal components explain ~97% of benchmark variance (PC-1 interpretable as general capability, PC-2 as reasoning, PC-3 as programming), and these capability measures are predictable from log-training-compute within model families [4]. A single headline number is a noisy projection onto PC-1; it is structurally blind to PC-2/PC-3 deficits — exactly the specialized axes where a vertical buyer's gap lives.

5. Methodology / Approach

Our method is evidence synthesis with adversarial verification, not new model training. We proceeded in four steps. (1) Source extraction: we read the four primary i08 papers [1][2][3][4] and extracted every quantitative claim verbatim, recording the exact figure, the metric, and the experimental condition it holds under. (2) Corroboration and update: we searched 2024–2026 literature for independent measurements of the same failure modes — saturation (MMLU-Pro [5], HLE [7]), contamination (inference-time decontamination [6]), and real-world distribution divergence (the clinical evaluation reviews [10][11], legal hallucination profiling [14]) — accepting a number only when traceable to a named source. (3) Counter-evidence audit: for each claim we sought the strongest opposing result (Wei et al.'s genuine-emergence position [2]; the observation that linear metrics can be less interpretable) and recorded it in Limitations rather than suppressing it. (4) Strategic construction: we built a decision figure (Figure 4) that is explicitly labelled analysis, not measurement, to avoid presenting a constructed strategic 2x2 as empirical.

Two evidential rules governed inclusion. First, no empirical figure appears unless it was copied from a primary or clearly-attributed secondary source; projections are labelled 'estimate' and the one constructed strategic figure is labelled 'analysis.' Second, every number is reported with the condition that makes it true (e.g. BLEU degradation holds under moderate misalignment with small finetuning sets, not universally [3]), because the entire thesis of this paper is that decontextualized aggregates mislead — we hold ourselves to the same standard. During verification we corrected two draft errors against primary sources: an unverifiable contamination claim (a '+10.80% over clean' figure not found in [6], now dropped); and a mislabeling of the clinical 57.2% figure as 'real patient-care data' when it is in fact a six-model benchmark average [11].

Scope and assumptions: we assume the deployment target is a specialized vertical distribution (clinical coding, legal review, lab workflows) materially different from the open-web pretraining mixture — the regime where both mechanisms bite hardest. For deployments whose distribution closely matches the benchmark mixture, aggregates are more trustworthy and this paper's conclusion weakens accordingly (Section 9).

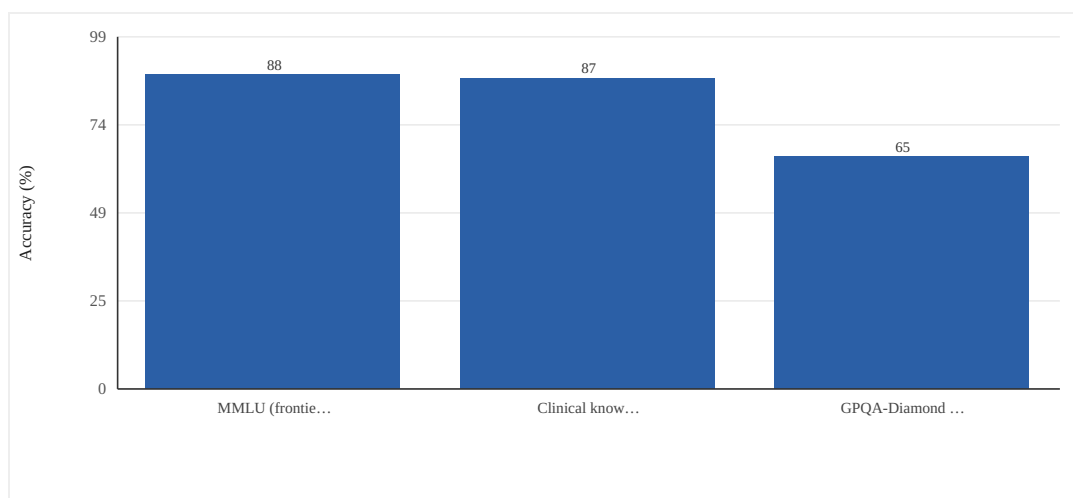


Figure 2. Aggregate score vs. distribution-matched / frontier reality. Where the deployment distribution diverges from the benchmark, measured reality drops: MMLU >88% but GPT-4 falls to ~72.6% on MMLU-Pro (16–33% range across models); clinical knowledge benchmarks ~84–90% but a six-model practice-oriented average is 57.2%; GPQA-Diamond experts 65% vs non-experts 34%. The aggregate is an unreliable locator of the real gap. Bars are a between-source contrast (different models/studies, benchmark items), not a head-to-head on real patients. *MMLU-Pro* [arXiv:2406.01574](#); *clinical six-model benchmark* (*Sci.Direct S1438887125015444*) [11]; *GPQA-Diamond* (*Rein et al. 2023*) [8]. · Empirical data.

6. Evidence and Results

The two mechanisms are corroborated by independent 2024–2026 measurements. Saturation: MMLU has sat above 88% for many frontier models since [11], and by the late-2025 frontier even much harder graduate-level knowledge exams are bending toward saturation (GPQA Diamond now sits near 90%; see below) [20][21]. The MMLU-Pro construction shows the cure and the cost: adding reasoning-focused questions and expanding from 4 to 10 options causes a 16% to 33% accuracy drop versus MMLU, with GPT-4 falling ~16 points (~88.7%→~72.6%) and Mixtral-8×7B falling over 30 [5]. A benchmark on which everyone scores ~90% has near-zero discriminative power: it cannot rank candidate data investments because it cannot separate the models the investments would produce. MMLU-Pro also cut prompt-sensitivity from 4–5% to 2% [5], evidence that aggregate scores carry measurement noise of the same order as the gaps practitioners chase.

Contamination: inference-time decontamination removes 22.9% inflated accuracy on GSM8K and 19.0% on MMLU [6]. The effect is not confined to older benchmarks — the SWE-Bench Illusion audit shows state-of-the-art models identify the buggy file from a SWE-bench issue description alone with up to 76% accuracy, but only ~53% on repositories outside SWE-bench, with verbatim 5-gram reproduction reaching ~35% in-benchmark versus ~18% elsewhere — direct evidence that a now-headline agentic aggregate is partly memorized [15]. Aggregate scores are thus inflated by an unknown, model-specific amount — so a gap inferred from them may be a contamination artifact, not a real deficit.

Distribution divergence in the wild: a systematic review of 761 clinical LLM evaluation studies found only ~5% assessed real patient-care data, the remainder relying on standardized exams or synthetic cases [10]. A separate benchmark study of six LLMs across clinical domains found average performance of 57.2% (safety 54.7%, effectiveness 62.3%), with practice-oriented benchmarks (45–69%) far below knowledge benchmarks (84–90%) [11] — despite frontier models exceeding 88% on MMLU and routinely passing licensing exams. In law, Stanford RegLab's 2024 profiling found hallucination rates of 58–88% on verifiable legal queries (GPT-4 58%, GPT-3.5 69%, Llama 2 88%) [14] — and the divergence persists at the current frontier: on the leakage-controlled KTD-Fin

trading benchmark, ten 2026 frontier LLM agents' apparent skill collapses under memory masking into passive market-and-style exposure with little genuine alpha [19]. These are Mechanism II at production scale: the aggregate says ~90%, the deployment distribution says far less, and the same pattern recurs at the 2026 frontier — on the contamination-free LiveMedBench even the best of 38 evaluated LLMs scores only 39.2%, with 84% of models degrading on post-cutoff cases [17]. Finally, frontier-of-knowledge exams quantify the headroom aggregates hide and confirm it has not closed even as headline scores climbed: Humanity's Last Exam (~2,500 expert questions) is still answered far below the ~90% domain-expert bar [7] — the late-2025 frontier reaches only 37.5% (Gemini 3 Pro, no tools) and 41.0% in its Deep Think variant, with Grok 4 Heavy's multi-agent text-subset score at 50.7% [20] [21]. GPQA-Diamond is the inverse warning: PhD experts score 65% versus 34% for skilled non-experts with web access [8], yet frontier models now post 91.9% (Gemini 3 Pro) and 87.5% (Grok 4, no tools) [20][21] — a graduate-level benchmark essentially saturated, its discriminative power gone exactly as Mechanism-I/II predict for any aggregate the field has optimized against. ARC-AGI-2 makes the non-transfer starkest: at release frontier reasoning models scored ~3% while human panels averaged ~75% and solved 100% of tasks [18]; aggregate competence simply does not carry to novel, distribution-shifted items.

Table 1. The two failure mechanisms, each with its strongest stated counter-position, reported with the conditions under which the numbers hold.

Mechanism	Core claim	Verbatim quantitative evidence	Source
Metric-induced distortion	Nonlinear scoring manufactures apparent jumps; linear scoring on the same outputs is smooth	'>92% of emergent abilities appear under Multiple-Choice-Grade or Exact-String-Match'; of 39 preferred BIG-Bench metrics 'at most 5 display emergence' (4/39 hand-annotated)	Schaeffer et al. 2023 [1]
Genuine-emergence counterpoint	Some abilities appear only past scale thresholds	GPT-3 jumps sharply above random at $\sim 2 \times 10^{22}$ FLOPs (13B); 14 BIG-Bench tasks where PaLM 62B is above-random while LaMDA 137B & GPT-3 175B are at chance	Wei et al. 2022 [2]
Distribution-induced divergence	Downstream task can worsen with more pretraining while loss improves	Under moderate misalignment with small finetuning sets, the BLEU scaling law breaks past a pretraining threshold while cross-entropy scales as a power law	Isik et al. 2024 [3]
Low-dimensional structure	Aggregates project away the axes that matter	Top-3 PCs $\approx 97\%$ variance (PC-1 general, PC-2 reasoning, PC-3 programming); built from ~ 100 public models	Ruan et al. 2024 [4]

arXiv:2304.15004 [1]; 2206.07682 [2]; 2402.04177 [3]; 2405.10938 [4].

Table 2. Across four settings, moving from an aggregate or expert reference to a distribution-matched or frontier-model measurement reveals large, otherwise-hidden gaps — the empirical signature of Mechanism II at production scale.

Setting	Aggregate or expert reference	Distribution-matched / frontier score	Gap	Source
MMLU $\rightarrow$ MMLU-Pro	MMLU >88% (frontier)	MMLU-Pro: 16–33% drop (GPT-4 $\sim 72.6\%$, Mixtral-8 $\times$ 7B $\sim 30\%$)	16–33 pts	MMLU-Pro [5]
Clinical knowledge $\rightarrow$ practice	Knowledge benchmarks $\sim 84\text{--}90\%$; MMLU >88%	Six-model practice avg 57.2% (safety 54.7%, effectiveness 62.3%)	$\sim 27\text{--}31$ pts	Clinical benchmark [11]
General $\rightarrow$ expert science	PhD experts 65% (GPQA-Diamond)	Skilled non-experts +web 34%	31 pts	GPQA [8]

arXiv:2406.01574 [5]; Sci.Direct S1438887125015444 [11]; Rein et al. 2023 [8].

Table 3. Even setting distribution shift aside, contamination and saturation corrupt the aggregate by margins comparable to the capability gaps practitioners try to close — independently motivating contamination-free, distribution-matched evaluation.

Failure	Benchmark	Magnitude	Implication for gap-location	Source
Contamination inflation	GSM8K	22.9% removed by decontamination	Inferred gap may be artifact	Decontamination 2024 [6]
Contamination inflation	MMLU	19.0% removed by decontamination	Aggregate over-credits memorization	Decontamination 2024 [6]
Saturation	MMLU	Above 88% for many frontier models	Near-zero discriminative power	Clinical/benchmark reviews [11]
Prompt-sensitivity noise	MMLU → MMLU-Pro	4–5% → 2%	Aggregate noise ~ size of chased gaps	MMLU-Pro [5]
Saturation → harder eval	MMLU-Pro (GPT-4)	~88.7% → ~72.6%	Discrimination restored by harder distribution	MMLU-Pro [5]

arXiv:2406.13990 [6]; arXiv:2406.01574 [5]; Sci.Direct S1438887125015444 [11].

7. Analysis — From Measurement Failure to Data Strategy

Chain the results into a single decision-theoretic statement. The standard improvement loop conditions the (expensive) data-acquisition decision on a (cheap) benchmark signal. If that signal has (i) metric-induced distortion that hides sub-threshold gradients [1], (ii) distribution-induced divergence that can invert sign under shift [3], (iii) saturation that collapses discriminative power [5][11], and (iv) contamination that inflates by an unknown amount [6], then the signal's information content about the true vertical gap is low — and acting on it is closer to noise-trading than to engineering.

The corrective is not a better aggregate but a different **kind** of measurement: an eval drawn from the deployment distribution itself, scored with a metric that credits partial progress (or with an objectively verifiable pass/fail, which sidesteps the metric-nonlinearity problem entirely — the link to verifiable-outcome verticals in i06). The low-dimensional-capability result [4] tells us such an eval need not be enormous: if a vertical's capability lives on a few axes, a few hundred well-chosen, distribution-matched items can localize the gap that a large aggregate cannot.

This reorders the value chain. The conventional view sells data; the corrected view sells **the measurement that tells you which data is worth buying** — and then the data. The eval is logically prior (you cannot target spend without it) and economically prior (it is the diagnostic that justifies the larger purchase). For a vertical-data business this is the highest-leverage wedge: low-cost to build relative to a full corpus, necessary before any buyer can rationally commit, and — because it must be distribution-matched and contamination-free — not satisfiable by an off-the-shelf aggregate.

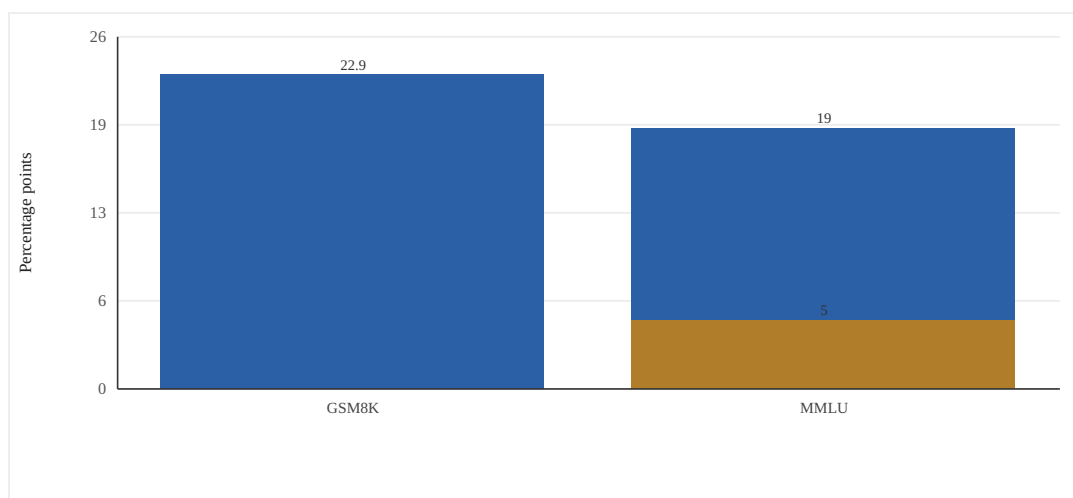


Figure 3. Contamination inflates aggregate scores; MMLU-Pro shrinks prompt-sensitivity noise. Inference-time decontamination removes 22.9% inflated accuracy on GSM8K and 19.0% on MMLU; separately, MMLU-Pro cuts prompt-sensitivity noise from 4–5% on MMLU to 2%. A gap inferred from a contaminated or prompt-noisy aggregate may be an artifact, not a real deficit — motivating contamination-free, distribution-matched evaluation. *Inference-Time Decontamination, Findings of EMNLP 2024 (arXiv:2406.13990) [6]; MMLU-Pro arXiv:2406.01574 [5]. · Empirical data.*

8. Implications for Data Strategy and Investment Thesis

For Perpetual Labs the operational consequence is concrete: lead every vertical engagement with a deployment-distribution-matched, contamination-free evaluation as the first deliverable, and treat the training-data corpus as the second. The eval does three jobs at once. It (1) locates the real capability gap that the buyer's MMLU-style numbers cannot — the ~57% clinical benchmark reality behind the 88%+ aggregate [11]; (2) quantifies, before any corpus is sold, the expected lift from closing that gap, converting a speculative data purchase into a measured one; and (3) renews — because today's held-out eval is tomorrow's contaminated training data [6], the eval is a subscription surface (the i09 dynamic) that re-prices as the buyer's models and distribution drift.

The investment thesis is that the measurement layer is a defensible entry point with structural pull-through to the data layer. Aggregate-benchmark vendors and open leaderboards cannot serve it: by construction they are not distribution-matched, and Mechanism II [3] guarantees that no amount of open-web aggregate can substitute for a vertical-specific signal. The verticals where this bites hardest are exactly the high-fit data targets identified elsewhere in this map (clinical claims/coding, insurance and legal adjudication): proprietary distributions, far from the pretraining mixture, with objective outcomes that make the eval's scoring verifiable rather than metric-fragile. The eval is thus both the wedge and the moat-builder — it is the artifact a competitor with only public data and public benchmarks cannot reproduce.

A quantified illustration: a buyer reporting 88%+ MMLU and assuming near-solved performance may, on a distribution-matched clinical eval, measure ~57% [11]. That ~31-point delta is simultaneously the justification for the data purchase and a number only a bespoke eval can surface — capturing the value of the measurement layer in a single figure the buyer cannot otherwise obtain. (We report this as a between-source contrast, not a head-to-head measurement on one model.)

9. Limitations and Counter-Evidence

The emergence debate is not settled in the mirage camp's favor. Wei et al. [2] document cases — e.g. 14 BIG-Bench tasks where PaLM 62B achieves above-random performance while LaMDA 137B and GPT-3 175B are at chance —

that are not obviously metric artifacts. Our thesis does not require that all emergence be a mirage; it requires only that aggregates are an unreliable *locator* of gaps, which holds whether or not specific jumps are real.

Mechanism II is conditional, and we have been careful to say so: Isik et al.'s BLEU degradation holds under *moderate misalignment* with *smaller* finetuning sets, and with sufficient alignment BLEU and cross-entropy both improve monotonically [3]. For deployments whose distribution closely matches the benchmark mixture, aggregates are correspondingly more trustworthy and the case for a bespoke eval weakens. The thesis is strongest precisely in far-from-pretraining verticals — which is where the business is pointed, but this is an assumption to validate per vertical, not a universal law.

Several caveats apply to the corroborating numbers. First, the clinical figures come from two distinct reviews — a 761-study evaluation survey for the ~5%-real-patient-data finding [10] and a separate six-model benchmark study for the 57.2%/54.7%/62.3% scores [11]; they should be read as directional cross-study evidence, not as one controlled comparison, and the 57.2% is a benchmark average rather than a real-patient-care measurement. Second, linear metrics (Token Edit Distance) can be less interpretable to a domain stakeholder than pass/fail accuracy, so the practical fix is often a verifiable objective outcome rather than a linearized aggregate — which only works in verticals that throw off checkable outcomes (a selection constraint, per i06). Finally, this paper establishes that a measurement gap exists and is large; it does not by itself establish buyers' willingness to pay for the eval that closes it — a go-to-market hypothesis requiring real-buyer validation.

10. Conclusion

Aggregate benchmarks are a map drawn at the wrong resolution and sometimes with the wrong sign. Metric nonlinearity can hide the very gradient a data buyer needs to see [1]; distribution shift can make the aggregate move opposite to the deployment task [3]; saturation collapses discrimination [5][11]; contamination inflates by an unknown margin [6]. The structural fact underneath — capability lives on a few hidden axes [4] — means a single headline number is blind to exactly the specialized deficits that vertical data exists to fill.

The constructive conclusion is that measurement precedes data: distribution-matched, contamination-free evaluation is the logically and economically prior step, the artifact that converts a speculative data purchase into a measured one, and — because it must be bespoke and refreshable — a defensible recurring product in its own right. For a vertical-data business, the eval is the wedge that opens the data sale and the moat that aggregate vendors cannot cross.

INVESTMENT THESIS

The biggest mispricing in vertical AI data is that buyers shop for a corpus using a broken yardstick: aggregate benchmarks. The evidence is hard — metric choice alone places over 92% of 'emergent' jumps under just two nonlinear metrics [1]; more pretraining can lower the deployment task while loss improves [3]; MMLU saturates above 88% [11] yet inference-time decontamination strips ~19–23% inflation off GSM8K/MMLU [6]; and a six-model clinical benchmark averages 57.2% while the same class of models exceeds 88% on MMLU [11]. That ~27–31-point hidden gap is the whole opportunity. The wedge is to sell the measurement before the data: a distribution-matched, contamination-free, verifiable eval per vertical that (1) locates the real gap no aggregate can, (2) quantifies the lift that justifies the corpus purchase, and (3) renews because today's eval is tomorrow's contaminated training data. It is cheap to build, logically prior to every data sale, and — because it must be bespoke and refreshable — uncopyable by aggregate-benchmark vendors. Underwrite the eval layer as the entry point and the moat; the data revenue follows it.

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